

Supporting Information (SI)

Life Cycle Assessment of Solvothermal ZIF-8 Synthesis: Is the Substitution of DMF with Glycerol Carbonate Environmentally Sustainable?

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S1. Glycerol carbonate synthesis and characterization

GlyC was synthesized through the transcarbonation of glycerol with dimethyl carbonate (DMC) using Na₂CO₃ as a catalyst. 517.07 g of DMC, 176.07 g of glycerol, and 0.61g of Na₂CO₃ were transferred into a 2L three-necked flask equipped with a magnetic stirrer (300 rpm) and subjected to heating at 100°C for 2h. Subsequently, the product was filtered to remove the catalyst, and the product was distilled under reduced pressure to remove the formed methanol and the excess of DMC. GlyC was obtained as a viscous liquid, detected through ¹H-NMR and ¹³C-NMR (Bruker AVANCE – 300 MHz using deuterated DMSO as the solvent), and the spectra are shown in Figure S1.1 and S1.2

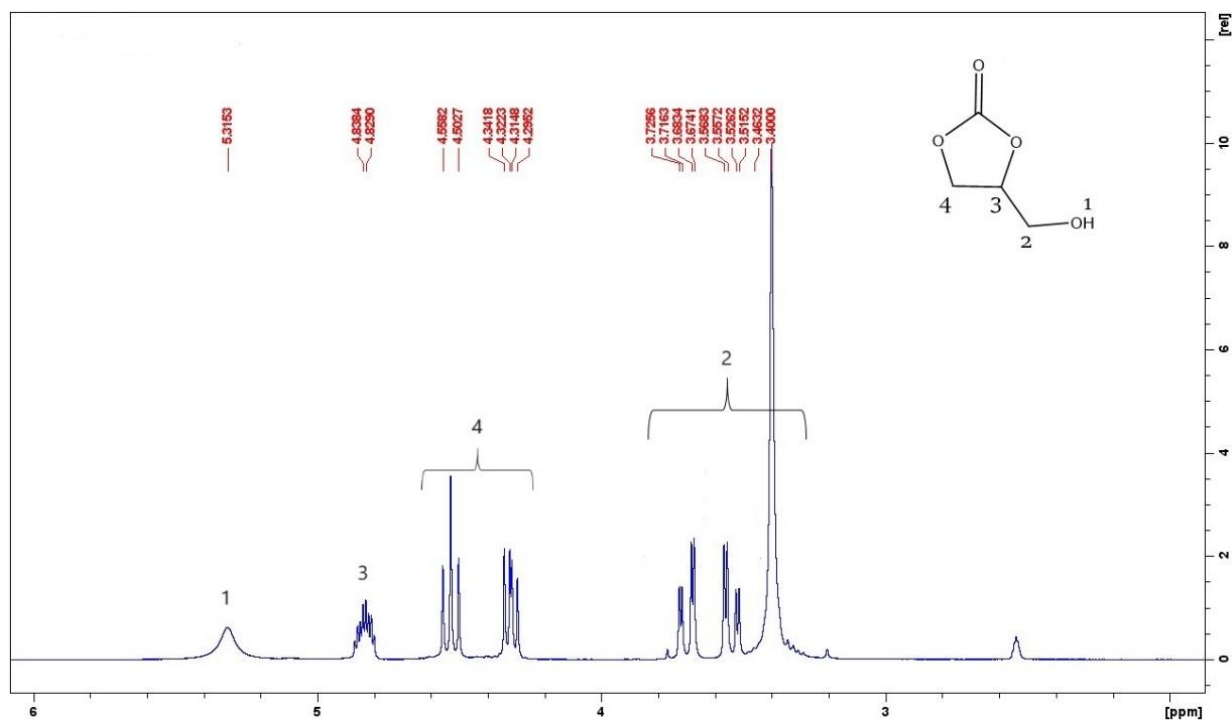


Figure S1.1 ¹H-NMR spectrum of Glycerol carbonate (GlyC) (300 MHz, DMSO).

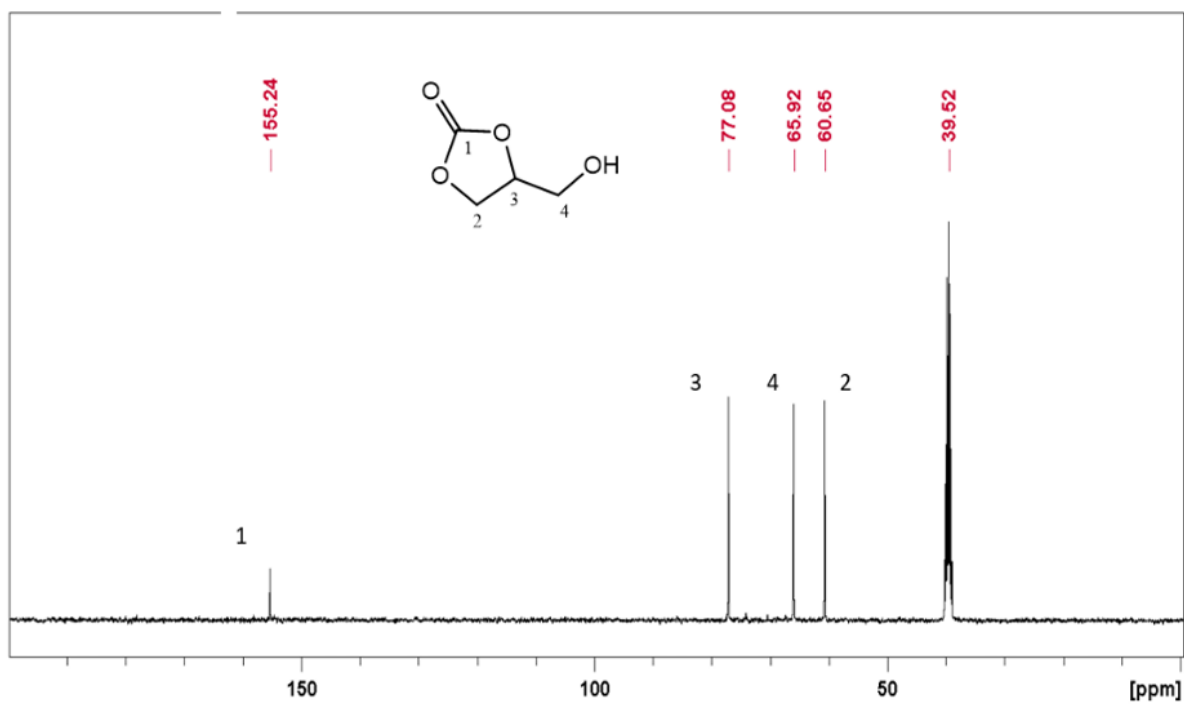


Figure S1.2. ¹³C-NMR of GlyC (100 MHz, DMSO), δ : 155.2 (CO), 77.1 (CH), 65.9 (CH₂), 60.6 (CH₂).

S2. Life Cycle Inventory (LCI)

S2.1 LCI of the main scenarios

Table S2.1.1 LCI of DMF-based Scenario (Z-DMF)

Item	Ecoinvent process	Type of process	Quantity	Unit of measurement
Zn (NO ₃) ₂ · 6H ₂ O	<i>_zinc nitrate 6 H₂O*</i>	INPUT	3.28	g
2-methylimidazole	<i>_2-Methylimidazole*</i>	INPUT	7.24	g
DMF	<i>N, N-dimethylformamide {GLO} market for APOS, U</i>	INPUT	51	g
Energy- ZIF-8 synthesis	<i>Electricity, medium voltage {GLO} market group for APOS, U</i>	INPUT	10.25	kWh
MeOH	<i>Methanol {GLO} market for APOS, U</i>	INPUT	20	g
Energy (Centrifugation)	<i>Electricity, medium voltage {GLO} market group for APOS, U</i>	INPUT	0.30	kWh
Energy (Drying)	<i>Electricity, medium voltage {GLO} market group for APOS, U</i>	INPUT	0.08	kWh
Waste to incineration	<i>spent solvent mixture {Europe without Switzerland} treatment of spent solvent mixture, hazardous waste incineration, with energy recovery APOS, U</i>	OUTPUT	60.39	g

Table S2.1.2 LCI of GlyC_baseline Scenario (Z-GlyC-base).

Item	Ecoinvent process	Type of process	Quantity	Unit of measurement
glycerol (RO)	<i>Glycerine {Europe without Switzerland} esterification of rape oil APOS, U</i>	INPUT	176.07	g
DMC	<i>Dimethyl carbonate {GLO} market for dimethyl carbonate APOS, U</i>	INPUT	517.07	g
Na ₂ CO ₃	<i>Soda ash, dense {GLO} market for APOS, U</i>	INPUT	0.61	g
energy for GlyC synthesis	<i>Electricity, medium voltage {GLO} market group for APOS, U</i>	INPUT	0.33	kWh
energy for GlyC distillation	<i>Electricity, medium voltage {GLO} market group for APOS, U</i>	INPUT	2.75	kWh
Waste to incineration <i>MeOH + DMC</i>	<i>spent solvent mixture {Europe without Switzerland} treatment of spent solvent mixture, hazardous waste incineration, with energy recovery APOS, U</i>	OUTPUT (waste)	(61+176)	g
Zn (NO ₃) ₂ · 6H ₂ O	<i>_zinc nitrate 6 H₂O (ZIF-8)</i>	INPUT	1.82	g
2-methylimidazole	<i>_2-Methylimidazole</i>	INPUT	1.36	g

Energy for ZIF-8 synthesis	<i>Electricity, medium voltage {GLO} market group for APOS, U</i>	INPUT	5.71	kWh
Energy (ultrasonic)	<i>Electricity, medium voltage {GLO} market group for APOS, U</i>	INPUT	0.15	kWh
MeOH	<i>Methanol {GLO} market for APOS, U</i>	INPUT	20	g
Energy (centrifugation)	<i>Electricity, medium voltage {GLO} market group for APOS, U</i>	INOPUT	0.30	kWh
Waste to incineration	<i>spent solvent mixture {Europe without Switzerland} treatment of spent solvent mixture, hazardous waste incineration, with energy recovery APOS, U</i>	OUTPUT (waste)	204.39	g

S2.2 GlyC-based optimized scenarios

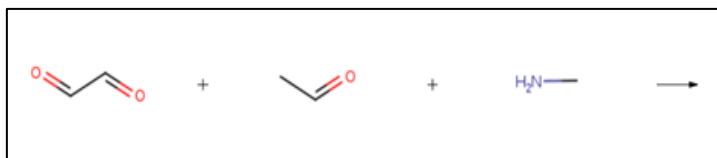
Mass balance of GlyC_baseline Scenario (Z-GlyC-wco): identical to Z-GlyC-base, it differs from this only for the origin of glycerol, Glycerine {FR}| treatment of waste cooking oil, purified, esterification | APOS, U, replacing “*Glycerine {Europe without Switzerland} | esterification of rape oil | APOS, U*”

Mass balance of GlyC_baseline Scenario (Z-GlyC-avp): in this scenario both *Dimethyl carbonate {GLO}| market for dimethyl carbonate | APOS, U* and *Methanol {GLO}| market for | APOS, U* appear as “**avoided product**” in the model.

Mass balance of GlyC_baseline Scenario (Z-GlyC-best): the improvements applied to Z-GlyC-wco and to Z-GlyC-avp were incorporated into this model.

The models for *_zinc nitrate*, *_zinc acetate*, *_2-methylimidazole* were designed as follows:

- *_2-methylimidazole*: It is prepared by condensation of glyoxal, ammonia, and acetaldehyde, nitration gives the 5-nitro derivative. Starting from “*Imidazole {RER}| production| APOS, U*” formaldehyde was replaced by acetaldehyde and ammonia was replaced by methylamine and calculations (all reagents’ quantities) were adjusted to be in agreement with the stoichiometry of the new reaction:



- a) 0.4644kg of formaldehyde (MW=30.031 g/mol) → 15,46 mol

$$\text{MW (acetaldehyde)} = 44.05 \text{ g/mol} \rightarrow w(\text{acetaldehyde}) = 15.46 \text{ mol} * \text{MW} = 0.681 \text{ kg.}$$

- b) 0.5263 kg of NH₃ (MW = 17.031 g/mol) → n = 30.9 mol

$$\text{MW CH}_3\text{NH}_2 = 31.05 \text{ g/mol, } w(\text{CH}_3\text{NH}_2) = 0.959 \text{ kg}$$

Item	Ecoinvent process	Type of process	Quantity	Unit of measurement
2-Methylimidazole	New	Product	1	kg
Water	Water, cooling, unspecified, natural origin, RER	INPUT	0.024	m ³
Methylamine	Methylamine {RER}, market for methylamine APOS, U	INPUT	0.959	kg
Chemical factory, organics {RER}	Construction APOS, U	INPUT	4x10 ⁻¹⁰	p
Acetaldehyde	Acetaldehyde {GLO} market for APOS, U	INPUT	0.681	kg
Glyoxal	Glyoxal {RER} market for glyoxal APOS, U	INPUT	0.898	kg

- zinc nitrate hexahydrate: $\text{ZnO} + 2 \text{HNO}_3 \rightarrow \text{Zn}(\text{NO}_3)_2 + \text{H}_2\text{O}$

Item	Ecoinvent process	Type of process	Quantity	Unit of measurement
zinc nitrate hexahydrate		product	1	g
Zinc oxide	zinc oxide {GLO} market for APOS, U	INPUT	0.4230	g
Nitric acid	nitric acid, without water, in 50% solution state {GLO} market for APOS, U	INPUT	0.665	g
Water	Water, deionized {EU-Switzerland}, water production, deionized APOS, U	INPUT	0.363	g

- zinc acetate dihydrate: **Current Patent Assignee:** RU2483056C2, 2011

Item	Ecoinvent process	Type of process	Quantity	Unit of measurement
zinc acetate dihydrate		product	1	g
Zinc oxide	zinc oxide {GLO} market for APOS, U	INPUT	0.370	g
Acetic acid	nitric acid, without water, in 98% solution state {GLO} market for APOS, U	INPUT	0.546	g
Water	Water, deionized {EU-Switzerland}, water production, deionized APOS, U	INPUT	0.164	g

S3. Z-DMF scenario

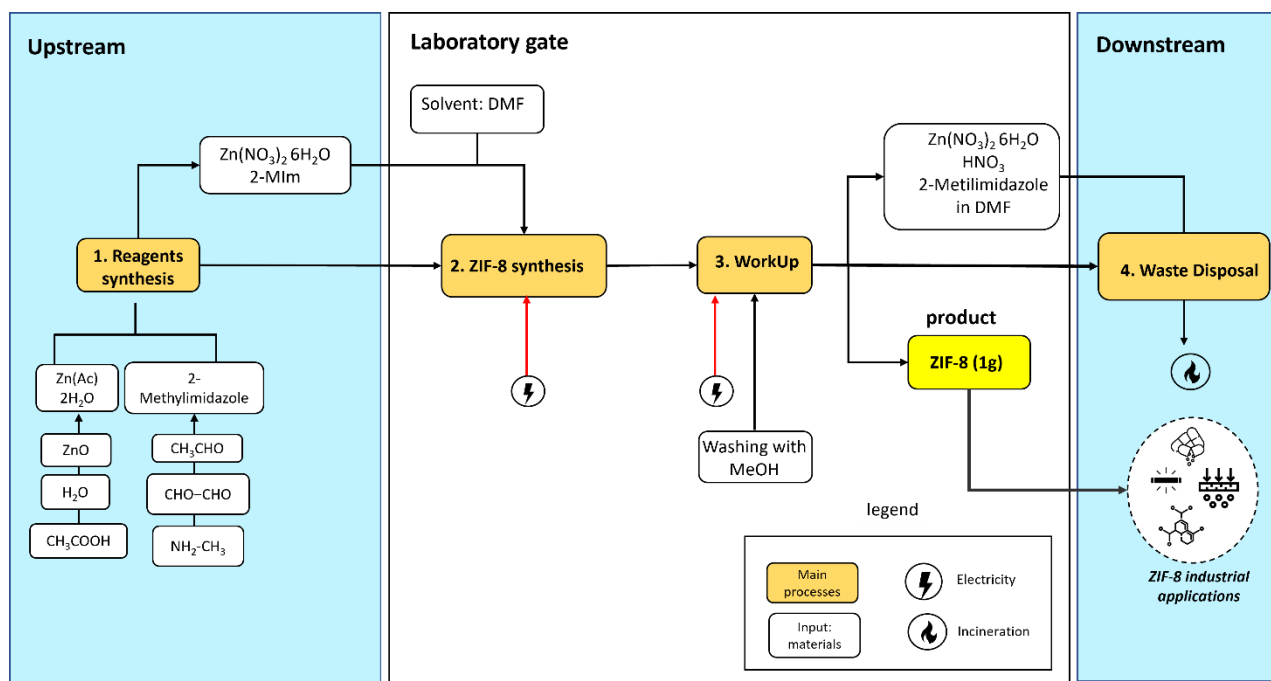


Figure S2.1 Flowchart of DMF-based Scenario (Z-DMF).

S4. Energy consumption data sheet

For all calculations, standard laboratory instrument power has been considered. Also, the following models for the instrument are: Hermle Labortechnik (centrifuge), Heidolph -MR Hei-Standard (heating plate), Leroy Somer (vacuum pump), DERUI DR-MH60 (Ultrasonic cleaner), GLOBEINSTRUMENTS (Oven).

For temperature allocation: power and time of usage were multiplied by the factor $\Delta t/\Delta t_{max} = (T_{reaction} - T_{ext})/(T_{max} - T_{ext})$, where $T_{ext} = T_{laboratory} = 19^{\circ}\text{C}$ ca.

Table S4.1 Energy requirement calculation for Z-DMF scenario ($T_{lab} = 19^{\circ}\text{C}$).

Convention oven	Unit	
Power	0.75	kW
Tmax	250	°C
ZIF-8 synthesis		
Treaction	140	°C
treaction	24	h
Work-up		
Treaction	120	°C
treaction	12	h

$\Delta T/\Delta t_{\max}$	0.57	
Speed _{max}	0.18	
Energy _(Synthesis)	10.25	kWh
Energy _(Essicc.)	0.08	kWh

Table S4.2 Energy requirement calculation for Z-GlyC scenario ($T_{\text{lab}} = 19\text{ }^{\circ}\text{C}$)

Parameter	Heating Plate (Heidolph MR Hei- Standard)	Vacuum pump (Leroy Somer)	Centrifuge (Hermle Labortechnik)	Ultrasonic (DERUI DR-MH60)
Power	0.825 kW	0.55 kW	0.58 kW	0.480 kW
Max Temperature (T_{max})	300 °C			
Max Speed (Speed_{max})	1400 rpm			

Parameter	GlyC-synthesis	GlyC Distillation	Work Up ZIF-8	Precursors' solutions
T reaction (°C)	75 °C	65°C		RT(Zn), 80°C (2-MIm)
Time	2 h	5h	20 min (tot.)	2 h(Zn), 1.5h(2-MIm)
Speed	250 rpm		5000 rpm	
$\Delta tT/\Delta T_{\max}$	0.20	0.20		
Speed/Speed Max	0.18	0.18	0.52	
Energy	0.33 kWh	2.75 kWh	0.30 kWh	0.378 kWh (Zn), 0.72(2-MIm)

ZIF-8 Synthesis

Parameter	T _{reaction}	Speed	Time	$\Delta T/T_{\max}$	Speed/Speed max	Energy
Value	100 °C	250 rpm	24 h	0.29	0.18	5.71 kWh

S5. Life Cycle Impact Assessment (LCIA)

Table S5.1.1 LCIA of the Z-DMF scenario ReCiPe 2016 Midpoint (H) V1.08 / World (2010) H/A.

Impact category	Unit	2-MIm (g)	Zn (NO ₃) ₂ (g)	DMF (g)	Electricity for mixing (kWh)	MeOH - Washing step (g)	Energy - Centrifugation (kWh)	Energy - Essiccation (kWh)	Waste (to incineration) (g)	total
Global warming	kg CO ₂ eq	4.87 x10 ⁻²	6.40 x10 ⁻³	1.56 x10 ⁻¹	7.66	1.39 x10 ⁻²	2.25 x10 ⁻¹	5.78 x10 ⁻²	1.19 x10 ⁻¹	7.96

Stratospheric ozone depletion	<i>kg CFC11 eq</i>	1.29 x10 ⁻⁸	1.13 x10 ⁻⁷	6.05 x10 ⁻⁸	3.18 x10 ⁻⁶	7.35 x10 ⁻⁹	9.34 x10 ⁻⁸	2.40 x10 ⁻⁸	3.16 x10 ⁻⁸	3.30 x10 ⁻⁶
Ionizing radiation	<i>kBq Co-60eq</i>	4.62 x10 ⁻³	1.20 x10 ⁻⁴	9.41 x10 ⁻³	9.27 x10 ⁻¹	2.23 x10 ⁻⁴	2.72 x10 ⁻²	6.99 x10 ⁻³	5.83 x10 ⁻⁴	9.45 x10 ⁻¹
Ozone formation, Human health	<i>kg NOx eq</i>	7.60 x10 ⁻⁵	9.40 x10 ⁻⁶	2.98 x10 ⁻⁴	1.67 x10 ⁻²	2.58 x10 ⁻⁵	4.90 x10 ⁻⁴	1.26 x10 ⁻⁴	4.79 x10 ⁻⁵	1.68 x10 ⁻²
Fine particulate matter formation	<i>kg PM2.5 eq</i>	4.79 x10 ⁻⁵	5.44 x10 ⁻⁶	2.08 x10 ⁻⁴	1.67 x10 ⁻²	1.30 x10 ⁻⁵	4.91 x10 ⁻⁴	1.26 x10 ⁻⁴	2.01 x10 ⁻⁵	1.69 x10 ⁻²
Ozone formation, Terrestrial ecosystems	<i>kg NOx eq</i>	7.74 x10 ⁻⁵	9.56 x10 ⁻⁶	3.03 x10 ⁻⁴	1.68 x10 ⁻²	2.63 x10 ⁻⁵	4.94 x10 ⁻⁴	1.27 x10 ⁻⁴	4.79 x10 ⁻⁵	1.68 x10 ⁻²
Terrestrial acidification	<i>kg SO2 eq</i>	1.44 x10 ⁻⁴	2.05 x10 ⁻⁵	4.92 x10 ⁻⁴	2.57 x10 ⁻²	3.57 x10 ⁻⁵	7.54 x10 ⁻⁴	1.94 x10 ⁻⁴	4.46 x10 ⁻⁵	2.60 x10 ⁻²
Freshwater eutrophication	<i>kg P eq</i>	1.60 x10 ⁻⁵	9.42 x10 ⁻⁷	5.73 x10 ⁻⁵	3.78 x10 ⁻³	1.65 x10 ⁻⁶	1.11 x10 ⁻⁴	2.85 x10 ⁻⁵	2.42 x10 ⁻⁵	3.88 x10 ⁻³
Marine eutrophication	<i>kg N eq</i>	2.64 x10 ⁻⁵	7.16 x10 ⁻⁸	1.56 x10 ⁻⁴	2.78 x10 ⁻⁴	1.32 x10 ⁻⁷	8.15 x10 ⁻⁶	2.10 x10 ⁻⁶	2.29 x10 ⁻⁶	4.55 x10 ⁻⁴
Terrestrial ecotoxicity	<i>kg 1,4-DCB</i>	4.63 x10 ⁻³	8.63 x10 ⁻²	7.83 x10 ⁻²	5.61	4.25 x10 ⁻⁴	1.65 x10 ⁻¹	4.23 x10 ⁻²	2.37 x10 ⁻³	1.24
Freshwater ecotoxicity	<i>kg 1,4-DCB</i>	6.09 x10 ⁻⁵	2.44 x10 ⁻⁴	3.58 x10 ⁻⁴	4.35 x10 ⁻¹	1.30 x10 ⁻⁵	1.28 x10 ⁻²	3.28 x10 ⁻³	3.32 x10 ⁻⁵	1.33 x10 ⁻²
Marine ecotoxicity	<i>kg 1,4-DCB</i>	6.54 x10 ⁻⁵	3.75 x10 ⁻⁴	2.76 x10 ⁻⁴	5.48 x10 ⁻¹	1.59 x10 ⁻⁵	1.61 x10 ⁻²	4.13 x10 ⁻³	4.82 x10 ⁻⁵	1.92 x10 ⁻²
Human carcinogenic toxicity	<i>kg 1,4-DCB</i>	4.86 x10 ⁻⁴	3.41 x10 ⁻⁴	2.43 x10 ⁻⁵	3.94 x10 ⁻¹	3.68 x10 ⁻⁶	1.16 x10 ⁻²	2.97 x10 ⁻³	1.88 x10 ⁻⁵	1.45 x10 ⁻³
Human non-carcinogenic toxicity	<i>kg 1,4-DCB</i>	3.90 x10 ⁻⁴	8.95 x10 ⁻³	1.44 x10 ⁻³	6.53	7.20 x10 ⁻⁵	1.92 x10 ⁻¹	4.92 x10 ⁻²	1.64 x10 ⁻⁴	6.06 x10 ⁻²
Land use	<i>m²a crop eq</i>	3.25 x10 ⁻³	2.40 x10 ⁻⁴	9.36 x10 ⁻³	3.22 x10 ⁻¹	3.25 x10 ⁻⁴	9.44 x10 ⁻³	2.43 x10 ⁻³	5.45 x10 ⁻⁴	3.40 x10 ⁻¹
Mineral resource scarcity	<i>kg Cu eq</i>	1.21 x10 ⁻⁴	1.93 x10 ⁻⁵	4.27 x10 ⁻⁴	7.84 x10 ⁻³	1.87 x10 ⁻⁵	2.30 x10 ⁻⁵	5.91 x10 ⁻⁵	3.52 x10 ⁻⁵	5.10 x10 ⁻³
Fossil resource scarcity	<i>kg oil eq</i>	2.57 x10 ⁻²	9.74 x10 ⁻⁴	8.10 x10 ⁻²	1.92	1.51 x10 ⁻²	5.64 x10 ⁻²	1.45 x10 ⁻²	3.85 x10 ⁻³	2.04
Water consumption	<i>m³</i>	9.09 x10 ⁻⁴	4.39 x10 ⁻⁵	1.97 x10 ⁻³	6.09 x10 ⁻²	8.45 x10 ⁻⁵	1.79 x10 ⁻³	4.59 x10 ⁻⁴	1.60 x10 ⁻⁴	6.05 x10 ⁻²

Table S5.1.2 LCIA of the Z-DMF scenario applying ReCiPe 2016 EndPoint (H) V1.08 / World (2010) H/A.

Damage-oriented category	Unit	2-MIM (g)	Zn (NO ₃) ₂ (g)	DMF (g)	Electricity for mixing (kWh)	MeOH – washing step (g)	Energy – Centrifugation step (kWh)	Waste (to incineration) (g)	total
Total	mPts	1.70	2.21 x10 ⁻¹	6.01	4.33 x10 ¹	4.52 x10 ⁻¹	5.38	2.26	3.56 x10 ⁻²
Human health	mPts	1.57	2.12 x10 ⁻¹	5.62	4.18 x10 ¹	4.00 x10 ⁻¹	5.20	2.15	3.44 x10 ⁻²
Ecosystem	mPts	6.10 x10 ⁻²	7.55 x10 ⁻³	1.92 x10 ⁻¹	1.16	1.46 x10 ⁻²	1.45 x10 ⁻¹	9.94 x10 ⁻²	9.70
Resources	mPts	6.60 x10 ⁻²	2.07 x10 ⁻³	1.95 x10 ⁻¹	3.12 x10 ⁻¹	3.81 x10 ⁻²	3.89 x10 ⁻²	1.02 x10 ⁻²	2.82

Values for energy (exsiccation step) appear to be negligible (=0).

Table S5.1.3. LCIA of the Z-DMF scenario applying Cumulative Energy Demand (CED) V1.11

Impact category	Unit	2-MIm(g)	Zn(NO ₃) ₂ (g)	DMF (g)	En. Mixing (kWh)	MeOH (washing step) (g)	Energy – Centrifugation (kWh)	waste (to incineration) (g)
Total	mPts	1.31	3.71 x10 ⁻²	3.99	1.18 x10 ⁻²	6.96 x10 ⁻¹	3.47	1.93 x10 ⁻¹
Non renewable, fossil	mPts	1.17	3.41 x10 ⁻²	3.70	8.83 x10 ¹	6.89 x10 ⁻¹	2.59	1.76 x10 ⁻¹
Non-renewable, nuclear	mPts	9.82 x10 ⁻²	1.69 x10 ⁻³	1.68 x10 ⁻¹	1.71 x10 ¹	3.97 x10 ⁻³	5.03 x10 ⁻¹	9.87 x10 ⁻³
Non-renewable, biomass	mPts	1.63 x10 ⁻⁵	2.87 x10 ⁻⁶	6.75 x10 ⁻⁵	8.22 x10 ⁻³	2.38 x10 ⁻⁶	2.41 x10 ⁻⁴	6.94 x10 ⁻⁶
Renewable, biomass	mPts	1.16 x10 ⁻²	4.92 x10 ⁻⁴	3.39 x10 ⁻²	2.08	5.30 x10 ⁻⁴	6.10 x10 ⁻²	1.84 x10 ⁻³
Renewable, wind, solar, geothermal	mPts	1.05 x10 ⁻²	1.94 x10 ⁻⁴	2.19 x10 ⁻²	3.13	5.03 x10 ⁻⁴	9.18 x10 ⁻²	1.34 x10 ⁻³
Renewable, water	mPts	1.89 x10 ⁻²	6.29 x10 ⁻⁴	6.06 x10 ⁻²	7.73	1.77 x10 ⁻³	2.27x10 ⁻¹	3.95 x10 ⁻³

Values for energy (exsiccation step) appear to be negligible (=0).

Table S5.2.1 LCIA of the Z-GlyC-base scenario ReCiPe 2016 MidPoint (H) V1.08 / World (2010) H/A.

Impact category	Unit	Glycerol RO (g)	DMC (g)	Na ₂ C O ₃ (g)	Electricity for mixing (kWh)	Energy Dist (kWh)	Waste to incineration (g)	Zn (OAc) ₂ (g)	Washing step	2Mim	Energy (Ultras) (kWh)	Energy (Ultras) (kWh)	NaOH(g)	Electricity for mixing (kWh)	Energy Centrifugation (kWh)	Incineration
GWP	kg CO ₂ eq	3.03 x10 ⁻¹	1.17	7.79 x10 ⁻⁴	2.46 x10 ⁻¹	2.06	4.68 x10 ⁻¹	2.24 x10 ⁻³	6.11 x10 ⁻²	9.15 x10 ⁻³	7.17 x10 ⁻²	3.77 x10 ⁻²	4.21 x10 ⁻⁴	4.27	2.25 x10 ⁻¹	4.04 x10 ⁻¹
SOD	kg CFC11 eq	4.75 x10 ⁻⁶	3.10 x10 ⁻²	2.02 x10 ⁻¹⁰	1.02 x10 ⁻⁷	8.54 x10 ⁻⁷	1.24 x10 ⁻⁷	1.07 x10 ⁻⁹	2.37 x10 ⁻⁸	2.43 x10 ⁻⁹	2.98 x10 ⁻⁸	1.56 x10 ⁻⁸	4.53 x10 ⁻¹⁰	1.77 x10 ⁻⁶	9.34 x10 ⁻⁸	1.07 x10 ⁻²
IR	kBq Co-60eq	1.33 x10 ⁻²	8.35 x10 ⁻²	3.65 x10 ⁻⁵	2.97 x10 ⁻²	2.49 x10 ⁻¹	2.29 x10 ⁻³	1.60 x10 ⁻⁴	3.69 x10 ⁻³	8.69 x10 ⁻⁴	8.68 x10 ⁻³	4.56 x10 ⁻³	4.66 x10 ⁻⁵	5.16 x10 ⁻¹	2.72 x10 ⁻²	1.97 x10 ⁻³
OF_HH	kg NO _x eq	9.32 x10 ⁻⁴	2.44 x10 ⁻³	1.47 x10 ⁻⁶	5.35 x10 ⁻⁴	4.48 x10 ⁻³	1.88 x10 ⁻⁴	5.56 x10 ⁻⁶	1.17 x10 ⁻⁴	1.43 x10 ⁻⁵	1.56 x10 ⁻⁴	8.21 x10 ⁻⁵	1.09 x10 ⁻⁶	9.29 x10 ⁻³	4.90 x10 ⁻⁴	1.62 x10 ⁻⁴

FPMF	kg PM2.5 eq	7.88 x10 ⁻⁴	1.39 x10 ⁻³	1.05 x10 ⁻⁶	5.37 x10 ⁻⁴	4.49 x10 ⁻³	7.90 x10 ⁻⁵	3.62 x10 ⁻⁶	8.17 x10 ⁻⁵	9.00 x10 ⁻⁶	1.57 x10 ⁻⁴	8.23 x10 ⁻⁵	9.08 x10 ⁻⁷	9.32 x10 ⁻³	4.91 x10 ⁻⁴	6.81 x10 ⁻⁵
OF_T E	kg NOx eq	9.38 x10 ⁻⁴	2.57 x10 ⁻³	1.48 x10 ⁻⁶	5.40 x10 ⁻⁴	4.51 x10 ⁻³	1.88 x10 ⁻⁴	5.87 x10 ⁻⁶	1.19 x10 ⁻⁴	1.46 x10 ⁻⁵	1.58 x10 ⁻⁴	8.27 x10 ⁻⁵	1.09 x10 ⁻⁶	9.37 x10 ⁻³	4.94 x10 ⁻⁴	1.62 x10 ⁻⁴
TA	kg SO2 eq	4.47 x10 ⁻³	3.04 x10 ⁻³	2.40 x10 ⁻⁶	8.24 x10 ⁻⁴	6.89 x10 ⁻³	1.75 x10 ⁻⁴	7.36 x10 ⁻⁶	1.93 x10 ⁻⁴	2.70 x10 ⁻⁵	2.41 x10 ⁻⁴	1.26 x10 ⁻⁴	1.53 x10 ⁻⁶	1.43 x10 ⁻²	7.54 x10 ⁻⁴	1.51 x10 ⁻⁴
FE	kg P eq	9.60 x10 ⁻⁵	3.50 x10 ⁻⁴	2.49 x10 ⁻⁷	1.21 x10 ⁻⁴	1.01 x10 ⁻³	9.48 x10 ⁻⁵	7.08 x10 ⁻⁷	2.25 x10 ⁻³	3.01 x10 ⁻⁶	3.54 x10 ⁻⁵	1.86 x10 ⁻⁵	2.14 x10 ⁻⁷	2.10 x10 ⁻³	1.11 x10 ⁻⁴	8.18 x10 ⁻⁵
ME	kg N eq	1.77 x10 ⁻³	3.19 x10 ⁻⁵	5.44 x10 ⁻⁷	8.91 x10 ⁻⁶	7.45 x10 ⁻⁵	9.00 x10 ⁻⁶	5.47 x10 ⁻⁸	6.11 x10 ⁻³	4.96x10 ⁻⁶	2.60 x10 ⁻⁶	1.37 x10 ⁻⁶	2.21 x10 ⁻⁸	1.55 x10 ⁻⁴	8.15 x10 ⁻⁶	7.76 x10 ⁻⁶
TET	kg 1,4-DCB	2.79 x10 ⁻²	1.04	5.47 x10 ⁻⁴	1.80 x10 ⁻¹	1.51	9.31 x10 ⁻³	4.31 x10 ⁻²	3.07 x10 ⁻²	8.71 x10 ⁻⁴	5.26 x10 ⁻²	2.76 x10 ⁻²	7.24 x10 ⁻⁵	3.13	1.65 x10 ⁻¹	8.03 x10 ⁻³
FET	kg 1,4-DCB	8.86 x10 ⁻⁴	1.64 x10 ⁻³	1.12 x10 ⁻⁶	1.40 x10 ⁻²	1.17 x10 ⁻¹	1.30 x10 ⁻⁴	1.32 x10 ⁻⁴	1.40 x10 ⁻⁴	1.14 x10 ⁻⁵	4.07 x10 ⁻³	2.14 x10 ⁻³	8.23 x10 ⁻⁷	2.42 x10 ⁻¹	1.28 x10 ⁻²	1.12 x10 ⁻⁴
MET	kg 1,4-DCB	8.32 x10 ⁻⁴	2.14 x10 ⁻³	1.53 x10 ⁻⁶	1.76 x10 ⁻²	1.47 x10 ⁻¹	1.89 x10 ⁻⁴	2.01 x10 ⁻⁴	1.08 x10 ⁻⁴	1.23 x10 ⁻⁵	5.13 x10 ⁻³	2.69 x10 ⁻³	1.20 x10 ⁻⁶	3.05 x10 ⁻¹	1.61 x10 ⁻²	1.63 x10 ⁻⁴
HCT	kg 1,4-DCB	4.60 x10 ⁻⁵	7.22 x10 ⁻⁴	1.52 x10 ⁻⁷	1.26 x10 ⁻²	1.06 x10 ⁻¹	7.37 x10 ⁻⁵	2.01 x10 ⁻⁴	9.52 x10 ⁻⁶	9.13 x10 ⁻⁵	3.69 x10 ⁻³	1.94 x10 ⁻³	1.53 x10 ⁻⁷	2.19 x10 ⁻¹	1.16 x10 ⁻²	6.36 x10 ⁻⁵
HNC T	kg 1,4-DCB	3.03 x10 ⁻³	1.40 x10 ⁻²	9.79 x10 ⁻⁶	2.09 x10 ⁻¹	1.75	6.45 x10 ⁻⁴	4.57 x10 ⁻³	5.65 x10 ⁻⁴	7.33 x10 ⁻⁵	6.12 x10 ⁻²	3.21 x10 ⁻²	6.45 x10 ⁻⁶	3.64	1.92 x10 ⁻¹	5.56 x10 ⁻⁴
LU	m2a crop eq	1.00	7.70 x10 ⁻²	5.07 x10 ⁻⁵	1.03 x10 ⁻²	8.63 x10 ⁻²	2.14 x10 ⁻³	1.48 x10 ⁻⁴	3.67 x10 ⁻³	6.10 x10 ⁻⁴	3.01 x10 ⁻³	1.58 x10 ⁻³	3.04 x10 ⁻⁵	1.79 x10 ⁻¹	9.44 x10 ⁻³	1.85 x10 ⁻³
MRS	kg Cu eq	1.13 x10 ⁻³	4.25 x10 ⁻³	3.27 x10 ⁻⁶	2.52 x10 ⁻⁴	2.10 x10 ⁻³	1.38 x10 ⁻⁴	8.30 x10 ⁻⁶	1.67 x10 ⁻⁴	2.28 x10 ⁻⁵	7.34 x10 ⁻⁵	3.86 x10 ⁻⁵	1.34 x10 ⁻⁶	4.37 x10 ⁻³	2.30 x10 ⁻⁴	1.19 x10 ⁻⁴
FRS	kg oil eq	6.40 x10 ⁻²	5.81 x10 ⁻¹	1.94 x10 ⁻⁴	6.17 x10 ⁻²	5.16 x10 ⁻¹	1.51 x10 ⁻²	1.17 x10 ⁻³	3.18 x10 ⁻²	4.83 x10 ⁻³	1.80 x10 ⁻²	9.46 x10 ⁻³	1.05 x10 ⁻⁴	1.07	5.64 x10 ⁻²	1.30 x10 ⁻²
WC	m3	1.50 x10 ⁻²	1.13 x10 ⁻²	1.41 x10 ⁻⁵	1.95 x10 ⁻³	1.63 x10 ⁻²	6.29 x10 ⁻⁴	4.72 x10 ⁻⁵	7.73 x10 ⁻⁴	1.71 x10 ⁻⁴	5.70 x10 ⁻⁴	2.99 x10 ⁻⁴	1.12 x10 ⁻⁵	3.39 x10 ⁻²	1.79 x10 ⁻³	5.42 x10 ⁻⁴

Table S5.2.2 LCIA of the Z-GlyC-*base* scenario ReCiPe 2016 EndPoint (H) V1.08 / World (2010) H/A.

Damage category	Unit	glycerol RO (g)	DMC (g)	Na ₂ CO ₃ (g)	Electricity for mixing(kWh)	Energy Dist(kWh)	Waste to incineration	Zn(OAc) ₂ (g)	2-mim (g)	EnergyUltras(kWh)	Energy - Uktras(kWh)	Washing step	Energy heating plate (kWh)	Energy Centrifugation(kWh)	Incineration
Human health	mPts	1.82 x10 ¹	4.18 x10 ¹	3.00 x10 ⁻²	1.10 x10 ¹	9.17 x10 ¹	8.45	1.03 x10 ⁻¹	2.96 x10 ⁻¹	3.20	1.68	2.21	1.90 x10 ²	1.00 x10 ¹	7.29
Ecosystem	mPts	2.96	1.45	9.95 x10 ⁻⁴	3.05 x10 ¹	2.55	3.90 x10 ¹	3.12 x10 ⁻³	1.15 x10 ⁻²	8.91 x10 ⁻²	4.68 x10 ⁻²	7.52 x10 ⁻²	5.30	2.79 x10 ¹	3.36 x10 ¹
Resources	mPts	1.60 x10 ⁻¹	1.48	3.73 x10 ⁻⁴	8.19 x10 ⁻²	6.85 x10 ⁻¹	3.99 x10 ⁻²	2.90 x10 ⁻³	1.24 x10 ⁻²	2.39 x10 ⁻²	1.26 x10 ⁻²	7.63 x10 ⁻²	1.42	7.49 x10 ⁻²	3.45 x10 ⁻²

Table S5.2.3 LCIA of the Z-GlyC- *base* scenario Cumulative Energy Demand (CED) V1.11

Impact category	Unit	Glycerol RO (g)	DMC (g)	Na ₂ CO ₃ (g)	Electricity for mixing (kWh)	Electricity for distillation(kWh)	Mixture to incineration	zinc acetate hexahydrate (g)	2-methylimidazole (g)	Energy sonication 2mim(kWh)	Energy sonication (kWh)	Energy heating plate (kWh)	Energy Centrifugation(kWh)	Incineration
Non renewable, fossil	MJ	2.92	2.65 x10 ¹	8.89 x10 ⁻³	2.83	2.37 x10 ¹	6.92 x10 ⁻¹	6.41 x10 ⁻²	2.20 x10 ⁻¹	8.27 x10 ⁻¹	4.34 x10 ⁻¹	4.92 x10 ¹	2.59	5.97 x10 ⁻¹
Non-renewable, nuclear	MJ	2.42 x10 ⁻¹	1.86	6.72 x10 ⁻⁴	5.50 x10 ⁻¹	4.60	3.87 x10 ⁻²	3.39 x10 ⁻³	1.85 x10 ⁻²	1.61 x10 ⁻¹	8.43 x10 ⁻²	9.55	5.03 x10 ⁻¹	3.34 x10 ⁻²
Non-renewable, biomass	MJ	8.80 x10 ⁻⁴	5.27 x10 ⁻⁴	3.71 x10 ⁻⁷	2.64 x10 ⁻⁴	2.20 x10 ⁻³	2.72 x10 ⁻⁵	2.75 x10 ⁻⁶	3.06 x10 ⁻⁶	7.69 x10 ⁻⁵	4.04 x10 ⁻⁵	4.57 x10 ⁻³	2.41 x10 ⁻⁴	2.35 x10 ⁻⁵
Renewable, biomass	MJ	7.57	2.99 x10 ⁻¹	2.04 x10 ⁻⁴	6.67 x10 ⁻²	5.58 x10 ⁻¹	7.22 x10 ⁻³	7.79 x10 ⁻⁴	2.17 x10 ⁻³	1.95 x10 ⁻²	1.02 x10 ⁻²	1.16	6.10 x10 ⁻²	6.22 x10 ⁻³
Renewable, wind, solar, geothermal	MJ	2.30 x10 ⁻²	1.95 x10 ⁻¹	8.59 x10 ⁻¹	1.00 x10 ⁻¹	8.40 x10 ⁻¹	5.24 x10 ⁻³	4.45 x10 ⁻⁴	1.97 x10 ⁻³	2.93 x10 ⁻²	1.54 x10 ⁻²	1.74	9.18 x10 ⁻²	4.52 x10 ⁻³
Renewable, water	MJ	1.03 x10 ⁻¹	5.08 x10 ⁻¹	3.01 x10 ⁻⁴	2.48 x10 ⁻¹	2.07	1.55 x10 ⁻²	1.39 x10 ⁻³	3.56 x10 ⁻³	7.24 x10 ⁻²	3.80 x10 ⁻²	4.31	2.27 x10 ⁻¹	1.34 x10 ⁻²

S6. Contribution analysis for Z-DMF

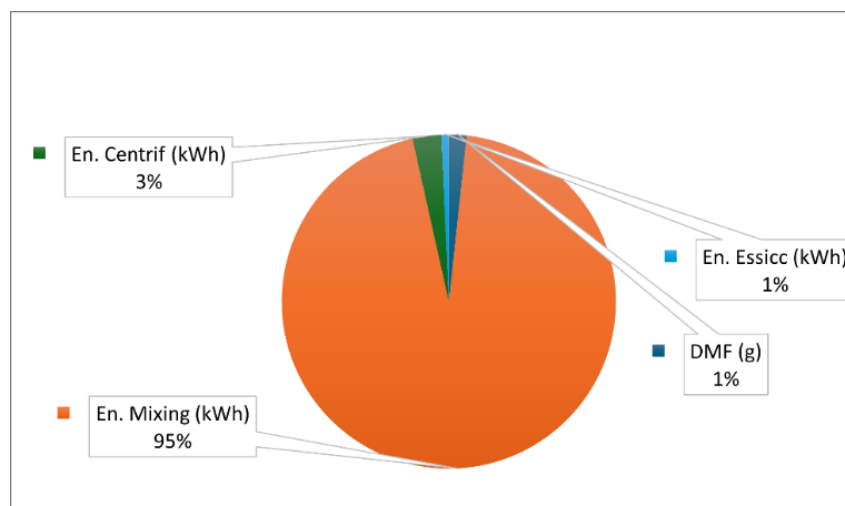


Figure S6.1 Contribution analysis expressed in cumulative single score through the ReCiPe EndPoint H approach for Z-DMF.

S7. Uncertainty analysis

Table S7.1 Pedigree Matrix value for both Z-DMF and Z-GlyC

Item	Reliability	Completeness	Temporal correlation	Geographical correlation	Further tecnological correlation
Z-DMF					
2-Mim	2	2	4	1	3
Zinc nitrate	2	2	4	1	4
DMF	2	2	4	1	1
Electricity - low voltage	2	2	4	1	4
Washing with MeOH	2	2	4	1	2
Z-GlyC					
2-Mim	2	2	4	1	3
Zn acetate	2	1	1	1	4
NaOH	2	1	1	1	1
glicerolo_rapeoil	1	1	1	1	1
glicerolo_wastecookingoil	1	2	1	2	1
DMC	1	1	1	1	1
Na2CO3	1	1	1	1	1
GlyC	2	4	1	1	4
Electricity - low voltage-GlyC synthesis	2	2	1	1	4
Incineration	2	2	1	1	2
Washing with MeOH	1	1	1	1	1
Electricity - GlyC distillation	2	2	1	1	4
Ultrasonic - electricity	2	2	1	1	1
Electricity for ZIF8 synthesis	2	2	1	1	4

Table S7.2 SD² values for both Z-DMF and Z-GlyC

Item	SD ² (σ^2)
Z-DMF	
<i>2-Mim</i>	1.2
<i>Zinc nitrate</i>	1.3
<i>DMF</i>	1.2
<i>Electricity - low voltage</i>	1.3
<i>Washing with MeOH</i>	1.2
Z-GlyC	
<i>2-Mim</i>	1.2
<i>Zn acetate</i>	1.2
<i>NaOH</i>	1.1
<i>glicerolo_rapeoil</i>	1.1
<i>glicerolo_wastecookingoil</i>	1.1
<i>DMC</i>	1.1
<i>Na₂CO₃</i>	1.1
<i>GlyC</i>	1.3
<i>Electricity - low voltage-GlyC synthesis</i>	1.2
<i>Incineration</i>	1.1
<i>Washing with MeOH</i>	1.1
<i>Electricity - GlyC distillation</i>	1.2
<i>Ultrasonic - electricity</i>	1.1
<i>Electricity for ZIF8 synthesis</i>	1.2

Monte Carlo analysis

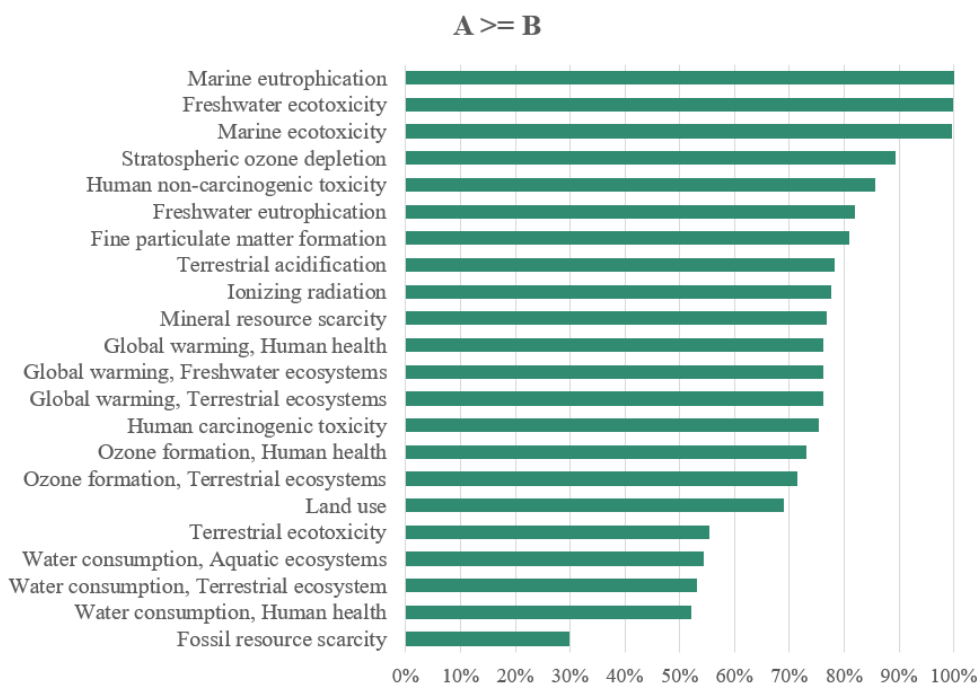


Figure S7.3 Uncertainty analysis of 1 g 'Z-DMF' (A) minus 1 g 'Z-GlyC-best' (B). Method: ReCiPe 2016 Endpoint (H) V1.05 / World (2010) H/A , confidence interval: 95 %

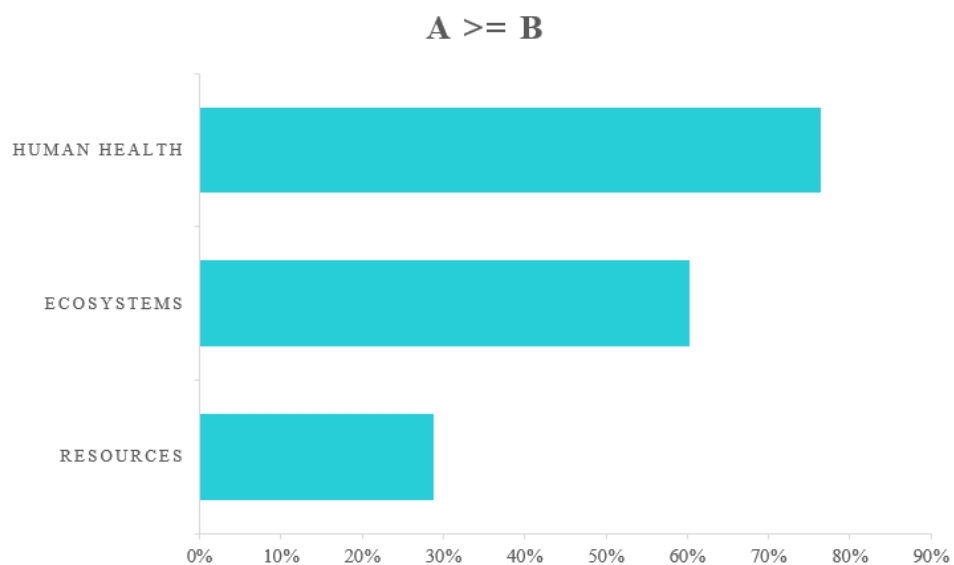
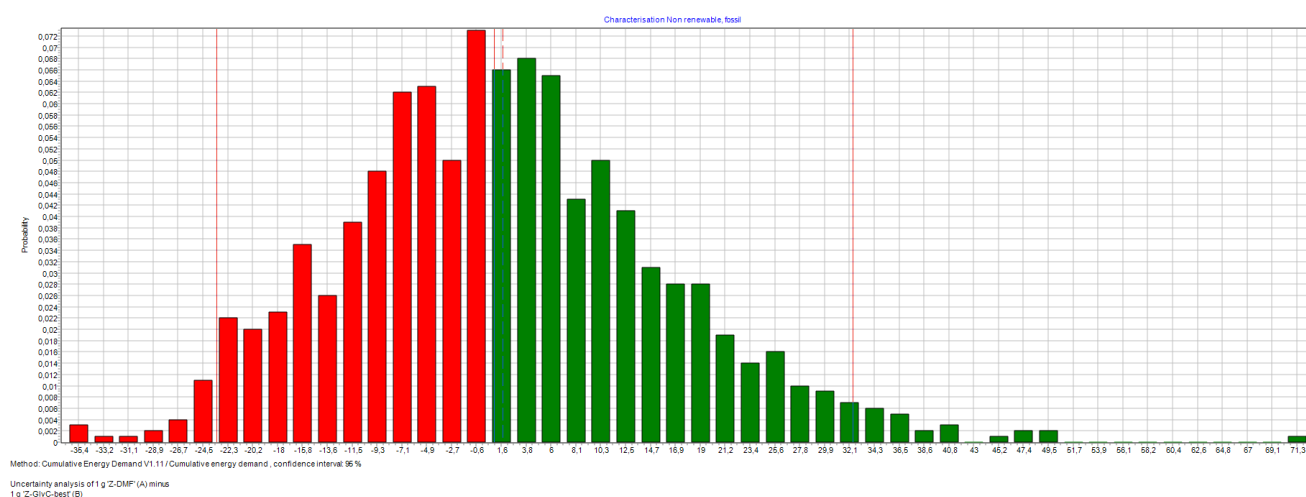


Figure S7.4 Uncertainty analysis of 1 g 'Z-DMF' (A) minus 1 g 'Z-GlyC-best' (B). Method: ReCiPe 2016 Endpoint (H) V1.05 / World (2010) H/A , confidence interval: 95 %



54.3% probability for A ≥ B

Figure S7.5 Uncertainty analysis of Z-DMF > Z-GlyC-best.

Method: Cumulative energy demand V1.1, confidence interval 95%.